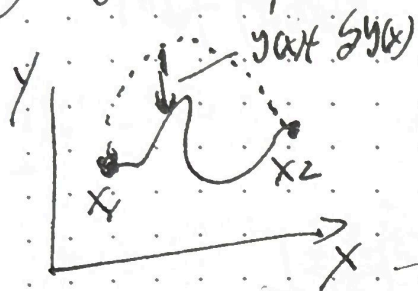


Variational Calculus:

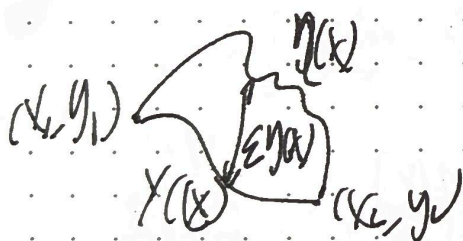
↳ (delta operator)



x_1, x_2 fixed.

~~$y(x) = \epsilon \eta(x)$~~
 $y(x) = \bar{y}(x) + \epsilon \eta(x)$
 saddle #

$$\delta[y(x)] = \bar{y}(x) - y(x) = \epsilon \eta(x)$$



$$\bar{y}(x) = y(x) + \epsilon \eta(x)$$

η : Delta
 η is the arbitrary variation of y , but must satisfy

$$\eta(x_1) = \eta(x_2) = 0$$

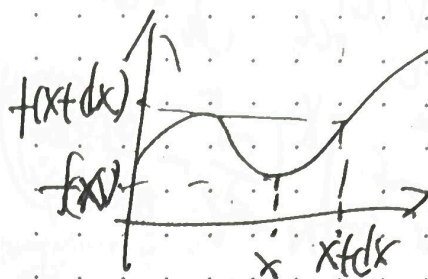
$$\bar{y}'(x) = y'(x) + \epsilon \eta'(x)$$

$$I = \int_{x_1}^{x_2} F(x, y(x), y'(x)) dx$$

$$\left. \frac{dI}{d\epsilon} \right|_{\epsilon=0} = \frac{d}{d\epsilon} \int_{x_1}^{x_2} F(x, y(x), y'(x)) dx = 0$$

Vs

Differential Operator



$$dy = f(x+dx) - f(x)$$



$x \neq x$ don't change much

$$\int_{x_1}^{x_2} \frac{d}{d\epsilon} \left(F(x, \bar{y}(x), \bar{y}'(x)) \right) \bigg|_{\epsilon=0} dx = 0$$

$$\int_{x_1}^{x_2} \left(\frac{\partial F}{\partial x} \frac{\partial x}{\partial \epsilon} + \frac{\partial F}{\partial y} \frac{\partial \bar{y}}{\partial \epsilon} + \frac{\partial F}{\partial y'} \frac{\partial \bar{y}'}{\partial \epsilon} \right) \bigg|_{\epsilon=0} dx = 0$$

$$\bar{y} = y(x) + \epsilon \eta(x)$$

$$\bar{y}'(x) = y'(x) + \epsilon \eta'(x)$$

$$\frac{\partial \bar{y}'}{\partial \epsilon} = \eta'(x)$$

$$\int_{x_1}^{x_2} \left(\frac{\partial F}{\partial y} \eta + \frac{\partial F}{\partial y'} \eta' \right) \bigg|_{\epsilon=0} dx = 0$$

$$@ \epsilon = 0, \bar{y} = y, \bar{y}' = y'$$

$$\int_{x_1}^{x_2} \left(\frac{\partial F}{\partial y} \eta + \frac{\partial F}{\partial y'} \eta' \right) dx = 0$$

1st variation
 weak form

$\bar{y}(x) \rightarrow$ non-optimized pass.
 (x_1, y_1) $\xrightarrow{\epsilon \eta(x)}$ (x_2, y_2)
 $y(x)$ is the optimized pass.

differential eqs

 Equations?

$\eta(x)$ arbitrary ϵ very small amplitude.

$$\bar{y}(x) = y(x) + \epsilon \eta(x), \quad \cancel{dy = dx + dy}$$

$\eta(x)$ is arbitrary, but $\eta(x_1) = \eta(x_2) = 0$
 it's called symmetric boundary condition.

If $f(x)$ is 'extremum':

$$I = \int_{x_1}^{x_2} F[x, y(x), y'(x)] dx$$

$$\left. \frac{d}{d\epsilon} F[x, \bar{y}(x), \bar{y}'(x)] \right|_{\epsilon=0} = 0 \quad \text{must have a stationary point}$$

$$\left. \frac{d}{d\epsilon} \int_{x_1}^{x_2} F[x, \bar{y}(x), \bar{y}'(x)] dx \right|_{\epsilon=0} = 0$$

Leibniz rule: $\int_{x_1}^{x_2} \frac{d}{d\epsilon} (F(x, \bar{y}(x), \bar{y}'(x))) \bigg|_{\epsilon=0} dx = 0$

need multivariable chain rule

$$\frac{\partial}{\partial \epsilon} F(x, y, y') = \frac{\partial F}{\partial x} \frac{\partial x}{\partial \epsilon} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial \epsilon} + \frac{\partial F}{\partial y'} \frac{\partial y'}{\partial \epsilon}$$

$$\int_{x_1}^{x_2} \left(\frac{\partial F}{\partial x} \frac{\partial x}{\partial \epsilon} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial \epsilon} + \frac{\partial F}{\partial y'} \frac{\partial y'}{\partial \epsilon} \right) \bigg|_{\epsilon=0} dx = 0$$

$$\int_{x_1}^{x_2} \left(\frac{\partial F}{\partial y} \eta(x) + \frac{\partial F}{\partial y'} \eta'(x) \right) dx = 0$$

1st variation
 weak form

Real: integration rule: $\int_a^b u dv = v u \Big|_a^b - \int_a^b v du$

$$\frac{\partial F}{\partial y} \eta \Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \left(\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right) \eta dx = 0 \quad \begin{array}{l} \text{2st variation} \\ \text{strong form} \end{array}$$

η is arbitrary

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

Euler-Lagrange
equation.